

Induced EMF of a Dropped Magnet

May 19, 2026

1 Objective

We drop a magnet through a coil from ten heights and read the peak-to-peak voltage V_{pp} off the scope each time, then plot V_{pp} versus drop height h and versus magnet speed at the coil $v = \sqrt{2gh}$.

2 Theory

Faraday's law gives the voltage induced across a coil of N turns by a changing flux Φ_B through one turn,

$$\varepsilon = -N \frac{d\Phi_B}{dt}. \quad (1)$$

For a magnet falling along the coil axis, the flux changes faster when the magnet moves faster, so the induced voltage is proportional to the magnet's speed v at the coil.

In free fall from release height h above the coil, $v = \sqrt{2gh}$, so $V_{pp} \propto v \propto \sqrt{h}$: V_{pp} vs. v should be linear through the origin, and V_{pp} vs. h should follow a square-root curve. The dominant uncertainty is the ± 0.5 cm operator placement of the magnet at release.

3 Apparatus

- Aluminum support rod (~ 1 m) C-clamped to the bench; a fixture at the top anchors a vertical guide line through the coil bore so the magnet falls along the coil axis.
- Small induction coil of insulated magnet wire on a bracket mounted to the rod at bench-top height, axis vertical, leads to scope ch 1 via a BNC adapter.
- Small permanent drop magnet; same magnet, same pole down each drop.
- Oscilloscope (Tektronix TDS 1002B, 60 MHz).
- Wooden meterstick clamped to the rod, reading h ; $\delta h = \pm 0.5$ cm from operator placement.

Circuit schematic

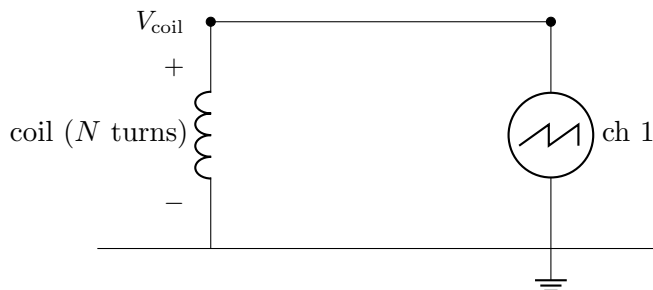


Figure 1: Circuit diagram. The coil terminals connect directly to oscilloscope channel 1; the scope reads the induced voltage $V_{\text{coil}}(t)$ as the magnet falls through the coil.

4 Procedure

1. Wire the coil to scope ch 1; set the trigger to Edge / CH1 / Rising / Normal / AC.
2. Clamp the meterstick vertically against the support rod with its zero at the coil top; use that reference for all ten trials.
3. Pick ten drop heights spanning roughly 10 cm to 100 cm. For each: place the bottom of the magnet at the target height, record $h \pm 0.5$ cm, release cleanly, freeze the trace with RUN/STOP, and read the peak-to-peak voltage $V_{\text{peak}} = V_{\text{max}} - V_{\text{min}}$ directly off the oscilloscope display by eye.

If a drop tumbles or the trace scrolls off-screen, re-take that height.

5 Data

At each of ten drop heights we released the magnet from rest and read $V_{\text{peak}} = V_{\text{max}} - V_{\text{min}}$ directly off the oscilloscope display. Each row of Table 1 is one trial. Heights span $h \approx 0.20$ m up to the bench-max $h \approx 0.96$ m (capped by the guide-line clearance above the coil). $\delta h = \pm 0.5$ cm is from operator placement, $\delta V_{\text{peak}} = \pm 0.04$ V from reading the display at the V/div used. Figure 2 plots the data against drop height with the square-root model overlaid.

Trial	h (m)	V_{peak} (V)
1	0.20 ± 0.005	1.12 ± 0.04
2	0.30 ± 0.005	1.20 ± 0.04
3	0.35 ± 0.005	1.28 ± 0.04
4	0.40 ± 0.005	1.36 ± 0.04
5	0.47 ± 0.005	1.40 ± 0.04
6	0.55 ± 0.005	1.44 ± 0.04
7	0.65 ± 0.005	1.52 ± 0.04
8	0.80 ± 0.005	1.84 ± 0.04
9	0.87 ± 0.005	1.92 ± 0.04
10	0.96 ± 0.005	2.12 ± 0.04

Table 1: Drop height h and peak-to-peak coil voltage V_{peak} for each of the ten drops.

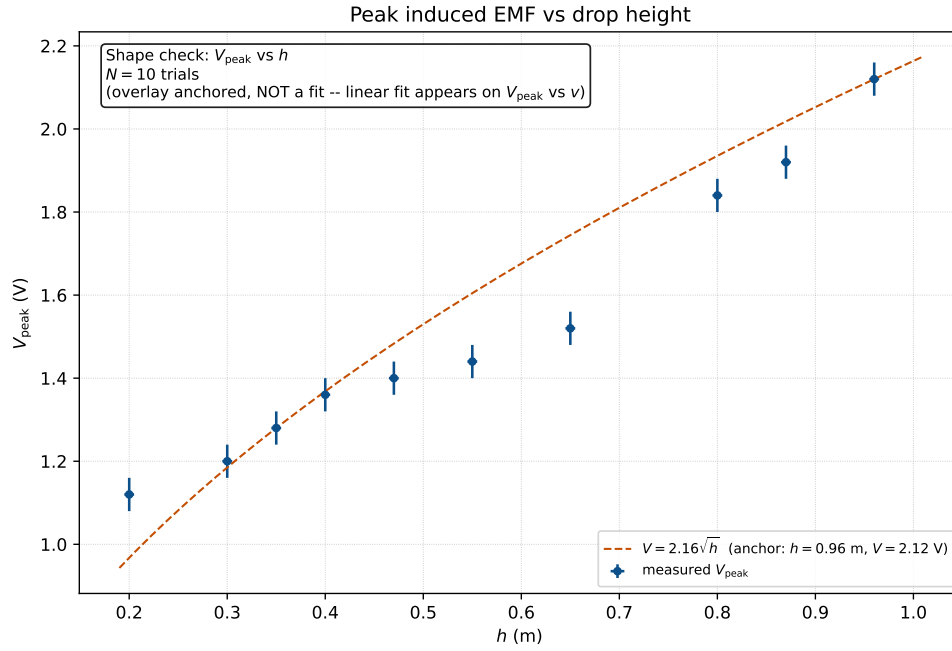


Figure 2: Peak-to-peak coil voltage V_{peak} versus drop height h , with the square-root model $V = K\sqrt{h}$ overlaid ($K \approx 2.16 \text{ V}/\sqrt{\text{m}}$, anchored at the bench-max point $h = 0.96 \text{ m}$, $V_{\text{peak}} = 2.12 \text{ V}$).

6 Calculations and Uncertainty

Propagation rules

$$\delta(A \pm B) = \sqrt{\delta_A^2 + \delta_B^2}$$

$$\frac{\delta(AB)}{|AB|} = \frac{\delta(A/B)}{|A/B|} = \sqrt{\left(\frac{\delta_A}{A}\right)^2 + \left(\frac{\delta_B}{B}\right)^2}$$

$$\delta f = \left| \frac{df}{dx} \right| \delta x$$

Velocity at the coil (Trial 6)

Inputs: $h = 0.55 \pm 0.005$ m, $g = 9.81$ m s⁻².

$$v = \sqrt{2gh}, \quad \delta v = \sqrt{\frac{g}{2h}} \delta h$$

$$v = \sqrt{2 \cdot 9.81 \cdot 0.55} = 3.285$$

$$\delta v = \sqrt{\frac{9.81}{2(0.55)}} \cdot 0.005 = 0.015$$

$$v = \boxed{3.29 \pm 0.02 \text{ m s}^{-1}}$$

The same propagation applied to every row of Table 1 gives the $v \pm \delta v$ values fed into the regression below.

Linear regression: $V_{\text{peak}} = m v + b$

A least-squares fit of V_{peak} versus $v = \sqrt{2gh}$ across all ten trials (Fig. 3) gives

$$m = \boxed{0.410 \pm 0.039 \text{ V/(m/s)}}$$

$$b = \boxed{0.203 \pm 0.128 \text{ V}}$$

$$R^2 = 0.933, \quad N = 10$$

The fit equation, R^2 , and N are annotated on Fig. 3.

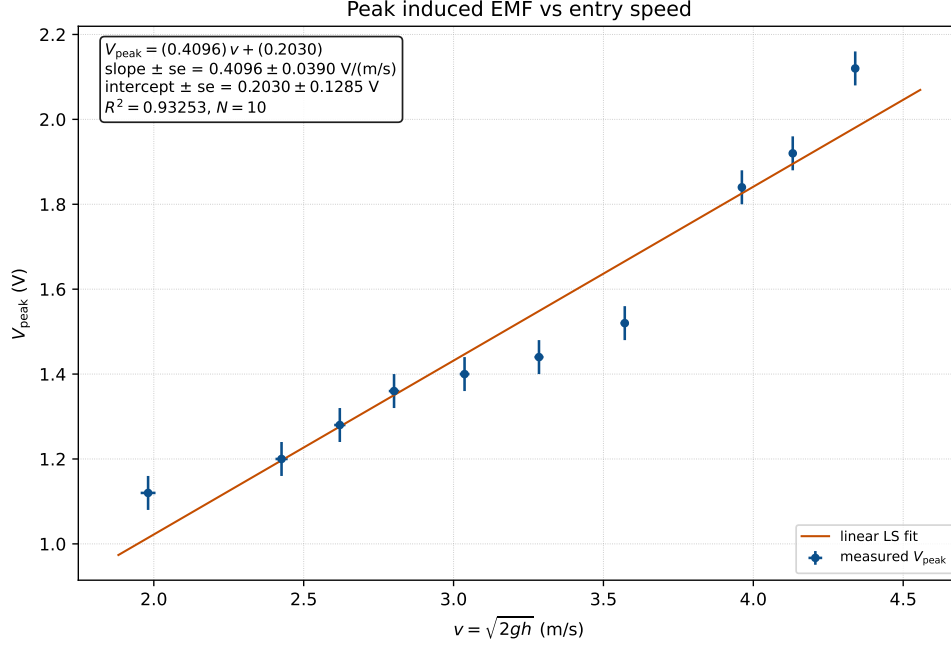


Figure 3: Peak-to-peak coil voltage V_{peak} versus magnet speed at the coil $v = \sqrt{2gh}$, with the least-squares linear fit. Faraday’s law predicts a straight line through the origin; the fit equation, R^2 , and N are shown on the figure.

t' test on the intercept

Faraday predicts strict proportionality, so the line should pass through the origin: $b = 0$.

$$t' = \frac{|b - 0|}{\delta b}$$

$$= \frac{0.203}{0.128}$$

$$t' = \boxed{1.58}$$

Since $1 \leq t' \leq 3$, the intercept sits in the *borderline* band: not consistent with zero at the 1σ level, but not in outright disagreement either. We cannot confirm the strict zero-intercept prediction from this dataset.

7 Conclusion

The peak-to-peak coil voltage V_{peak} is linear in the magnet’s speed at the coil $v = \sqrt{2gh}$, with fitted slope 0.41 ± 0.04 V/(m/s), intercept 0.20 ± 0.13 V, and $R^2 = 0.93$ over ten drops. The slope is statistically nonzero ($t' = 10.5$), confirming the qualitative Faraday prediction $V \propto v$. The intercept is small but not quite consistent with zero ($t' = 1.58$, *borderline*); Faraday predicts a strict zero intercept, and a *borderline*-nonzero fit reads most simply as residual reading error or scope baseline drift rather than a violation of the physics.

The companion $V_{\text{peak}}\text{-vs-}h$ plot shows the same data with a $\sqrt{h}$ overlay anchored at the bench-max point ($h = 0.96\text{ m}$, $V_{\text{peak}} = 2.12\text{ V}$); the data tracks the square-root shape qualitatively. The scatter is moderate ($R^2 = 0.93$ rather than $0.99+$), consistent with one trial per height and manual display reads; a visible kink near $h \approx 0.8\text{ m}$, where V_{peak} jumps from 1.52 V (trial 7) to 1.84 V (trial 8), carries most of the residual.

The dominant uncertainty is the $\pm 0.5\text{ cm}$ operator placement on h , which propagates through $v = \sqrt{2gh}$ into $\delta v \approx 0.016\text{ m s}^{-1}$ and on into the slope's standard error.